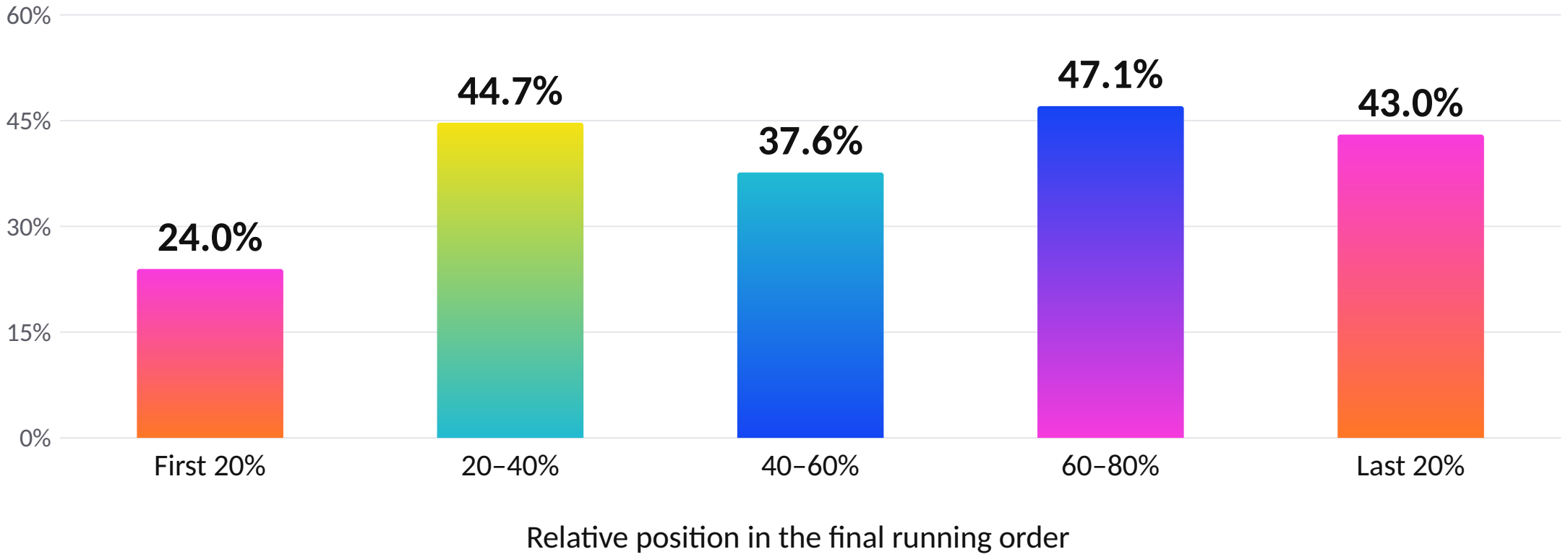
CAN WE FORECAST EUROVISION?

Top 10, audience vs jury, national televote winners

Nikita Nazarov Ivan Tanaev

Early performers reach the Top 10 less often

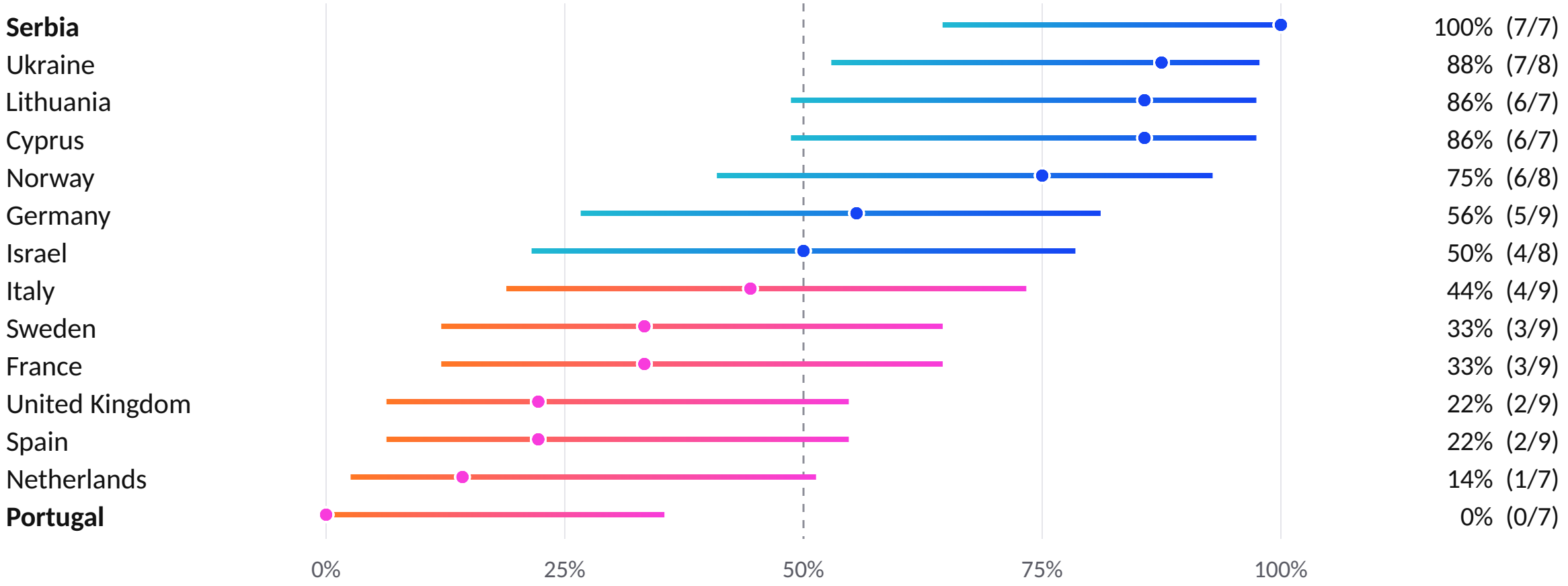
2008–2025 · 437 finalists · observed Top 10 frequency



Observed frequencies. Descriptive association, without adjustment for other song characteristics.

Audience and jury preferences differ by country

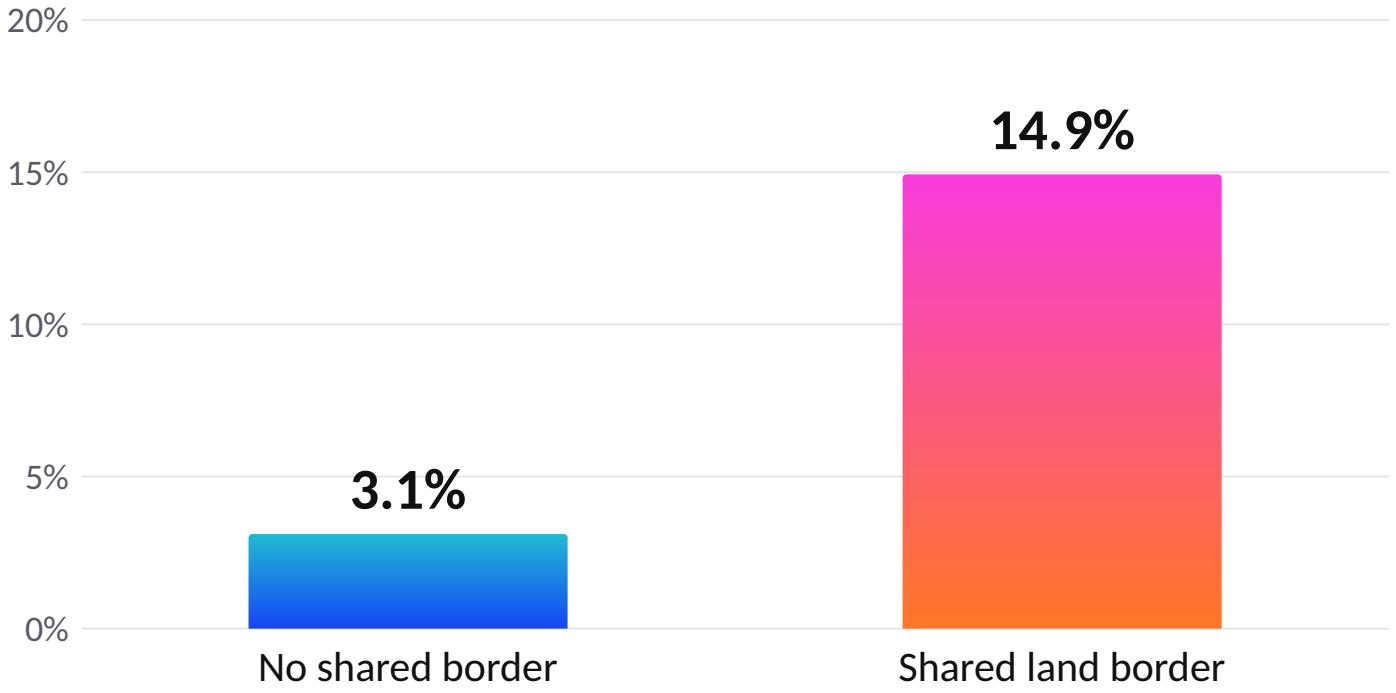
2016–2025 · countries with at least seven finals



Share of finals with audience rank better than jury rank. Lines: 95% Wilson intervals, assuming independent years.

Neighbours more often win each other’s televote

2016–2025 · genuine national televotes · eligible voter–finalist pairs



4.8×

the observed frequency for neighbouring pairs

8,127 pairs without a border
643 pairs with a border

Frequency of a national first place. Descriptive association, without adjustment for song popularity.

Voting preferences have a direction

2016–2025 · selected country pairs

Voter	RU	BY	UA	MD	AM	AZ	GE	EE	LV	LT	GR	CY	RO	ME	RS	PL	SI	HR
Russia		0/1	0/3	1/2	1/1	1/3	0/1	0/2	0/1	0/3	0/2	1/4				0/1	0/2	0/1
Belarus	2/2		1/3	0/2	0/2	0/3	0/1	0/2	0/1	0/2	0/2		0/1		0/3	0/2	0/2	0/2
Ukraine	1/2	0/1		1/5			0/2		0/3	1/7			0/2			2/5	0/3	
Moldova	3/3	0/2	3/7				0/2		0/2				1/2					
Armenia	2/2	0/2					0/2	2/6	0/3			2/6	0/2					
Azerbaijan	2/3	0/2	1/8				0/2		0/3				0/2					1/4
Georgia	0/3	0/2	2/8		3/6	1/5			0/3		1/6	1/7	0/2					
Estonia	2/3	0/2	2/8				0/2		0/3	1/7			0/2					
Latvia	2/3	0/2	1/8				0/2	2/6		2/7			0/2					
Lithuania	1/3	0/2	3/8				0/2	1/6	2/3				0/2					
Greece	0/3	0/2					0/2		0/3			7/7	0/2					
Cyprus	0/3	0/2	1/8				0/2		0/3		5/6		0/2					
Romania	0/2	0/2		5/5	0/3											0/3	0/3	0/2
Montenegro	0/2	0/2		0/3			0/1		0/2				0/2		4/4		0/2	1/2
Serbia	1/3	0/2		1/5			0/2	1/6	0/3				0/2					1/4
Poland	0/3	0/2	7/8				0/2		0/3				0/2					
Slovenia	0/3	0/2					0/2		0/3				0/2		4/7			3/4
Croatia	0/3	0/2					0/2	1/6	0/3				0/2		5/7		1/4	

ROWS: country casting the vote
COLUMNS: country receiving the vote

7 / 7

Greece voting for Cyprus

5 / 5

Romania voting for Moldova

Blue / orange: more / less often than other voters choose the same country.
Grey: no eligible votes. Blank coloured cell: zero wins.

7/7 means seven first places out of seven occasions when this vote was possible.

What goes into the models

12 predictors for A and B · 14 for C · the same predictors in M1 and M2

Block	Predictors	Tasks
Performance	Running position, host country	A / B / C
Song	Language, tempo, danceability, dynamics, word count, repetitiveness	A / B / C
History	Past country performance or past support within each country pair	A / B / C
Geography	Distance between capitals, shared land border	C
Market	Log probability of overall victory from a fixed panel of five bookmakers	A / B / C

Song Science · Eurovision results · Natural Earth · Eurovisionworld

Historical predictors use earlier contests. Archived market snapshots have no verified timestamp.

Data coverage

Finalists from 2008 · split results and national televotes from 2016

437
finalist entries

259
entries with market data

232
audience–jury outcomes

349
national voting events

	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Finalists / song features	25	25	25	25	26	26	26	27	26	26	26	26	×	26	25	26	25	26
Audience vs jury	–	–	–	–	–	–	–	–	26	26	26	26	×	26	25	26	25	26
National televotes	–	–	–	–	–	–	–	–	41	41	42	40	×	38	39	36	36	36
Market probabilities	–	–	–	–	–	–	–	27	26	26	26	26	×	26	25	26	25	26

2020 was cancelled. Counts refer to finalists, except the national-televote row.

Market coverage: 2015–2025, excluding 2020. C contains 8,770 candidate rows grouped into 349 voting events.

Each forecast uses earlier contests

Expanding-window evaluation · 2025 is held out from all training

	2015–19	2021	2022	2023	2024	2025
Forecast 2021	Train	Evaluate				
Forecast 2022	Train	Train	Evaluate			
Forecast 2023	Train	Train	Train	Evaluate		
Forecast 2024	Train	Train	Train	Train	Evaluate	
Forecast 2025	Train	Train	Train	Train	Train	**Final test**

A / B: Brier score + binary log loss

C: categorical log loss + Brier

Lower is better. Compare models on the same contests, with equal weight per year.

Initial training: A uses 2015–2019; B/C use 2016–2019. Scaling is fitted on training data only.

A. A model of the complete ranking

Plackett–Luce · one ranking generates both win and Top 10 probabilities

1st → **2nd** → **… Nth**

Choose among all finalists → Choose among those remaining → Continue until everyone is ranked

$$\Pr(i \text{ is next} \mid R) = \frac{\exp(\eta_i)}{\sum_{j \in R} \exp(\eta_j)}$$

R denotes remaining finalists; utility measures how strongly the model favours each finalist.

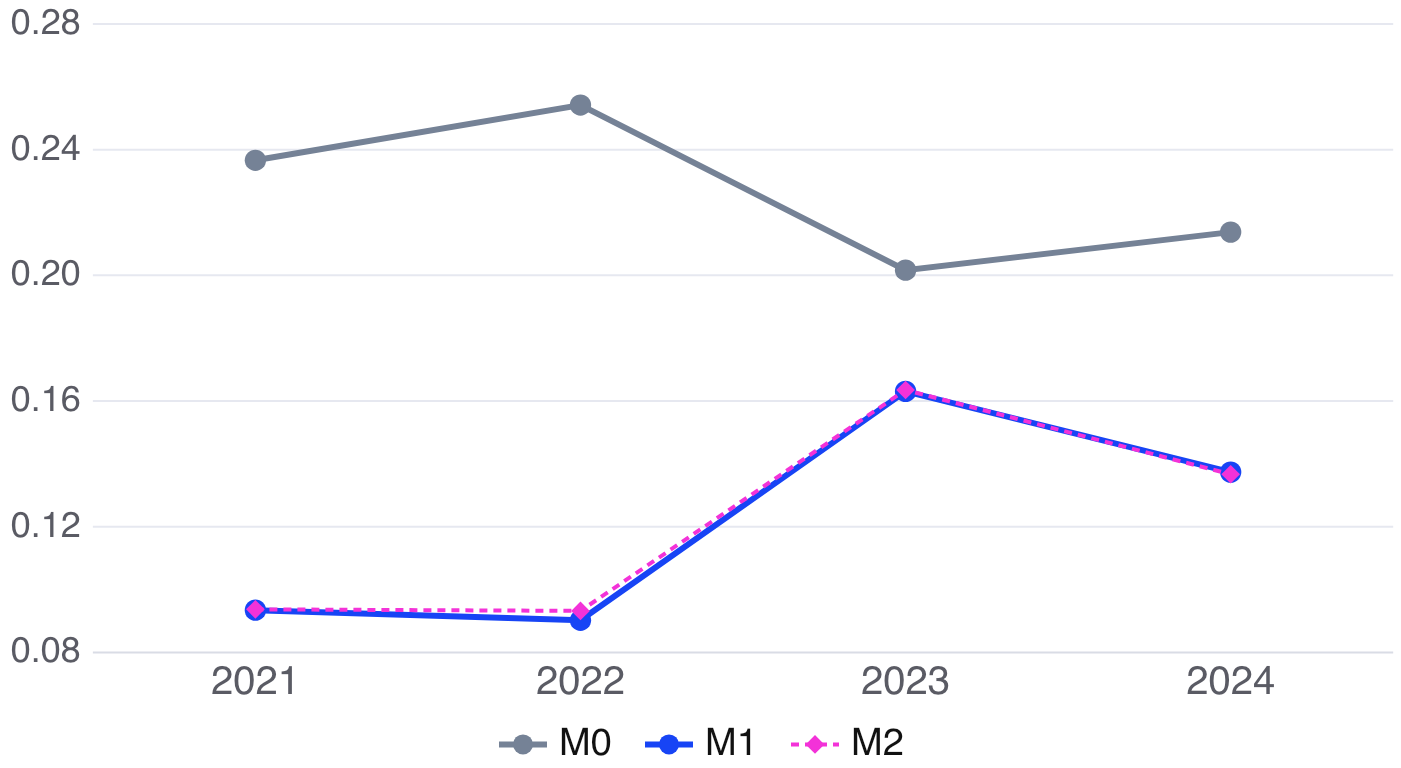
Model	Utility	Country effects
M0	$\eta_i = a_{c(i)}$	Historical strength · fixed prior scale
M1	$\eta_i = a_{c(i)} + \mathbf{x}_i^\top \beta$	Same country prior + 12 predictors
M2	$\eta_i = a_{c(i)} + \mathbf{x}_i^\top \beta$	Learned country scale · partial pooling

Simulated rankings contain exactly one winner and ten Top 10 places. Predictors are the same in M1 and M2.

A. Predictors improve Top 10 forecasts

2021–2024 validation · 2025 evaluated separately · market predictors included

Brier score · lower is better



Top 10 · Brier score

Model	2021–24 mean	2025
M0	0.2266	0.2036
M1	**0.1210**	**0.1519**
M2	0.1218	0.1519

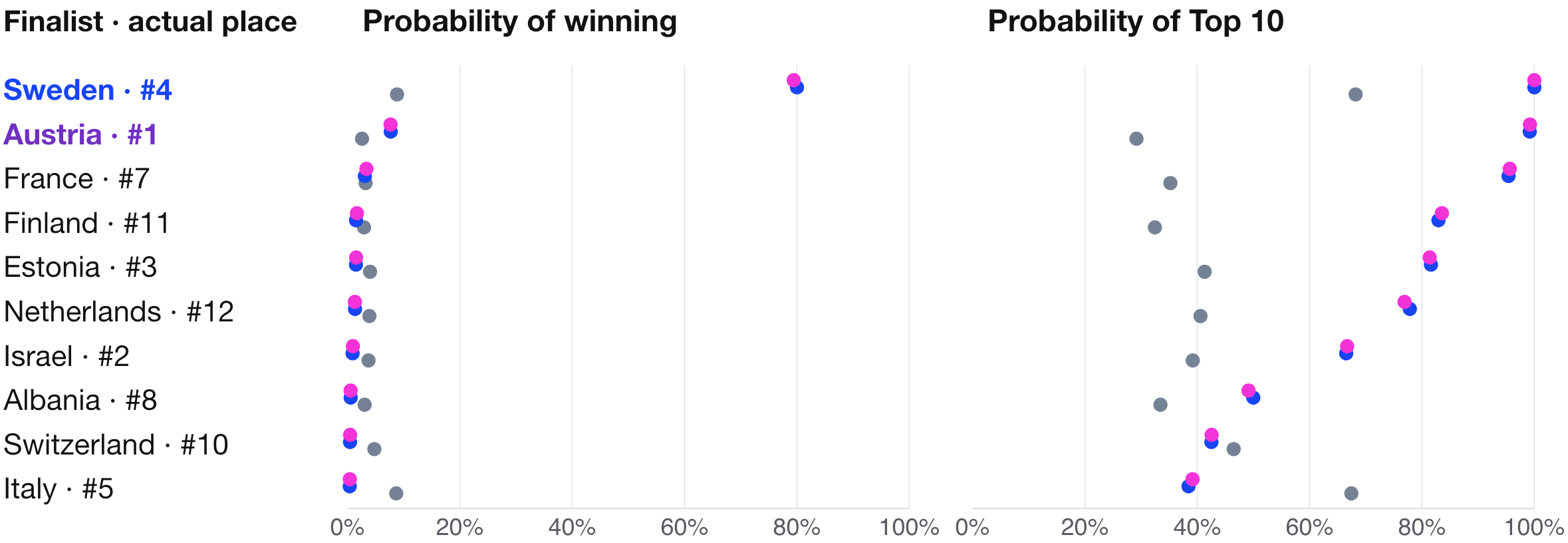
47%

lower mean Brier for M1 than M0 in 2021–2024

M1 and M2 are nearly tied. Four validation years, with equal weight per year. Results are retrospective.

A. Sweden was the favourite for 2025

Austria ranked second by win probability in M1/M2 — and won the contest



M1 win probability: Sweden 80.0% · Austria 7.7%

Ten highest Top 10 probabilities in M2 shown; all 26 finalists were modelled. Labels show actual overall places.

B. Will the audience rank a country higher?

One binary event per finalist · a lower rank number means a better result

$$Y_i = \mathbf{1}\left\{ r_i^{\text{audience}} < r_i^{\text{jury}} \right\}$$

1 if the audience rank is better
0 otherwise, including ties

We compare ranks, not the number of points

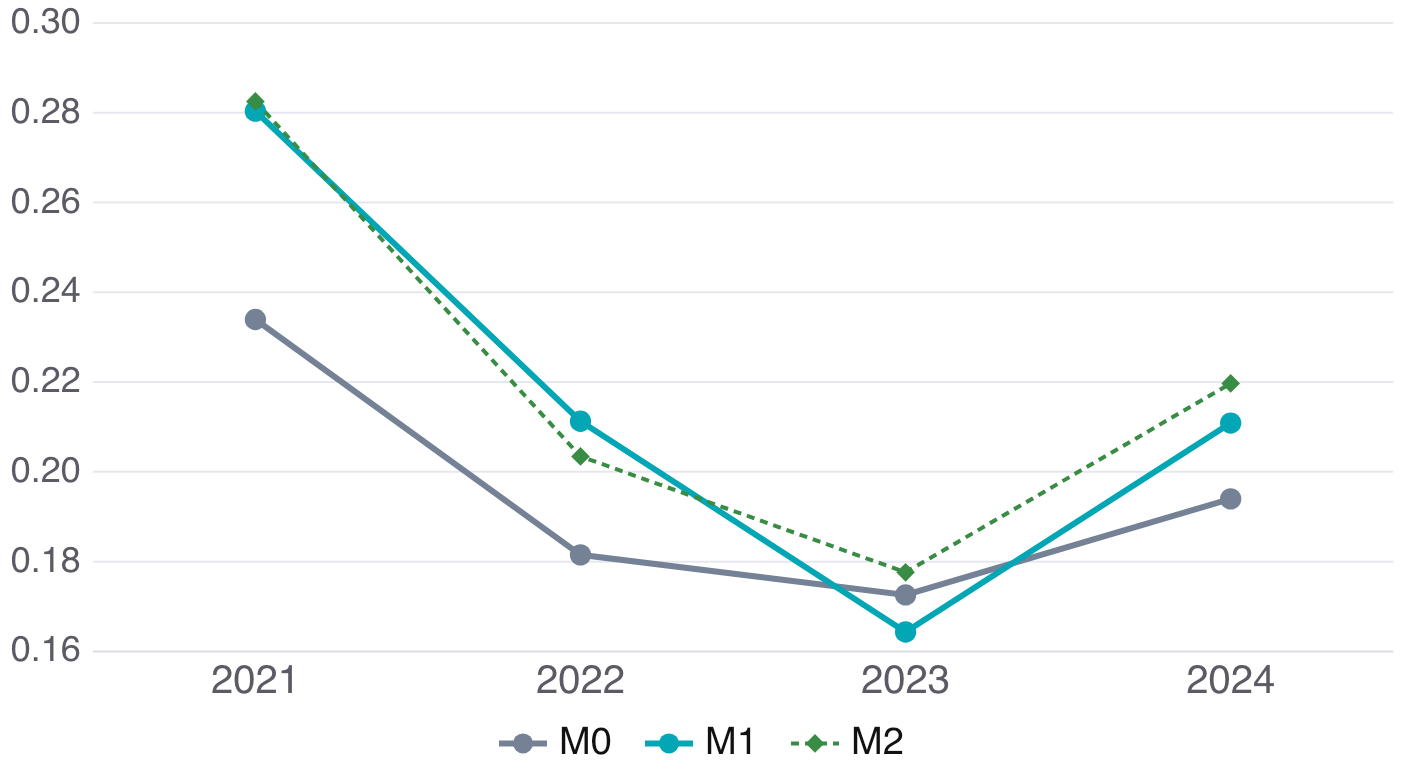
Model	Probability / latent score	What changes
M0	$\theta_c \sim \text{Beta}(1,1), \qquad p_c = \dfrac{s_c + 1}{n_c + 2}$	Smoothed history of each country
M1	$\Pr(Y_i = 1 \mid \eta_i) = \Phi(\eta_i), \qquad \eta_i = a_{c(i)} + \mathbf{x}_i^{\mathsf{T}}\beta$	12 predictors + fixed country prior
M2	$\eta_i = \alpha + a_{c(i)} + \mathbf{x}_i^{\mathsf{T}}\beta, \qquad a_c = \tau(z_c - \bar{z})$	Same predictors + hierarchy and an overall intercept

M0: Beta–Bernoulli; successes and appearances are counted separately for each country, using earlier contests only.

B. Country history beats extra predictors

2021–2024 validation · 2025 evaluated separately · market predictors included

Brier score · lower is better



Audience vs jury · Brier score

Model	2021–24 mean	2025
M0	**0.1955**	**0.1598**
M1	0.2167	0.1733
M2	0.2208	0.1733

M0

best mean Brier and log loss
in validation and in 2025

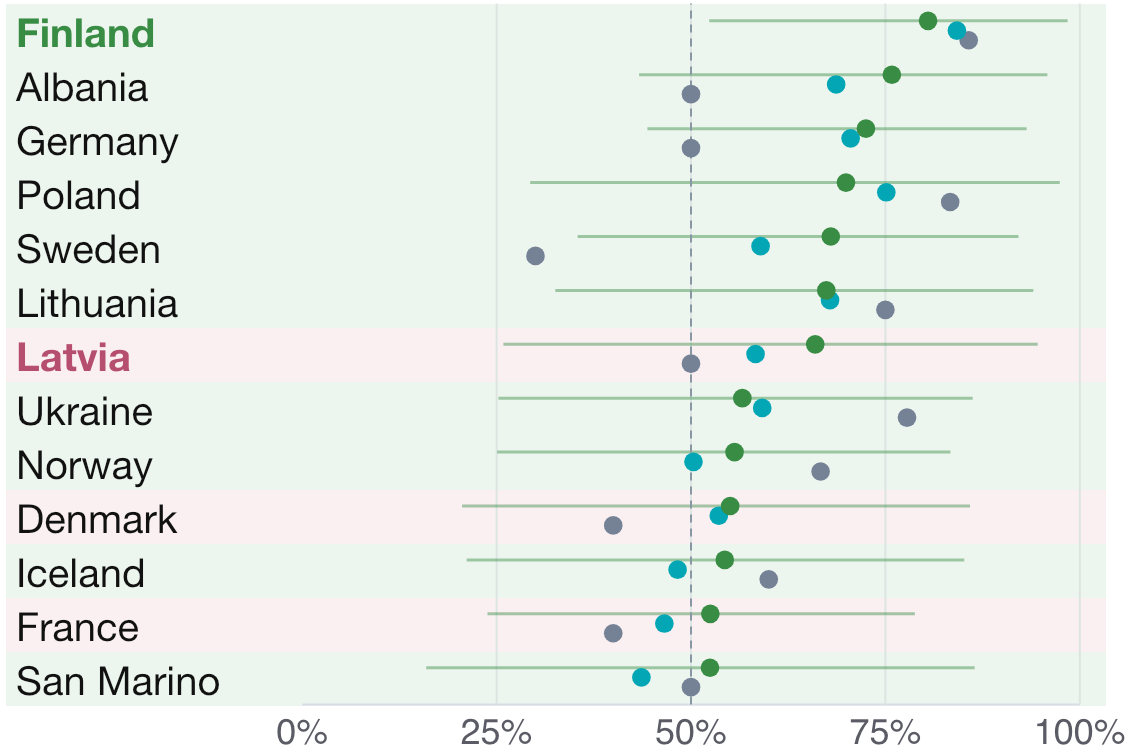
Extra predictors did not improve average accuracy here. Four validation years, with equal weight per year.

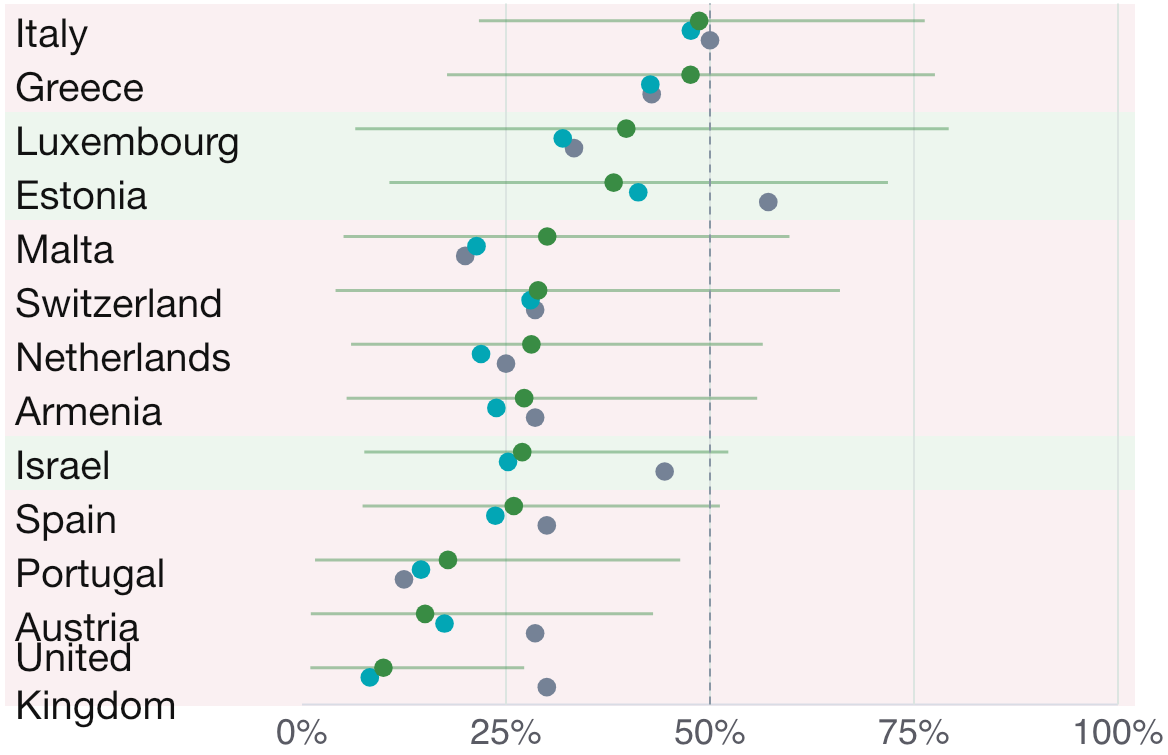
B. 2025 forecasts and observed outcomes

Probability that audience rank is better than jury rank · all 26 finalists

M0 M1 M2

Green rows: audience higher · pink rows: otherwise





Finland: M2 80%; audience ranked higher

Latvia: M2 66%; audience did not rank higher

Bars: 90% credible intervals for M2 probabilities. Dashed line: 50%. Ordered by M2; read down each column.

C. Who will each country give 12 points to?

Softmax over eligible finalists · one winner of each national televote

$$\Pr(W_{vt} = r \mid \theta) = \frac{\exp(\eta_{vrt})}{\sum_{j \in C_{vt}} \exp(\eta_{vjt})}$$

Probabilities sum to 1 within each national vote

No self-voting. Countries outside the final can still vote.

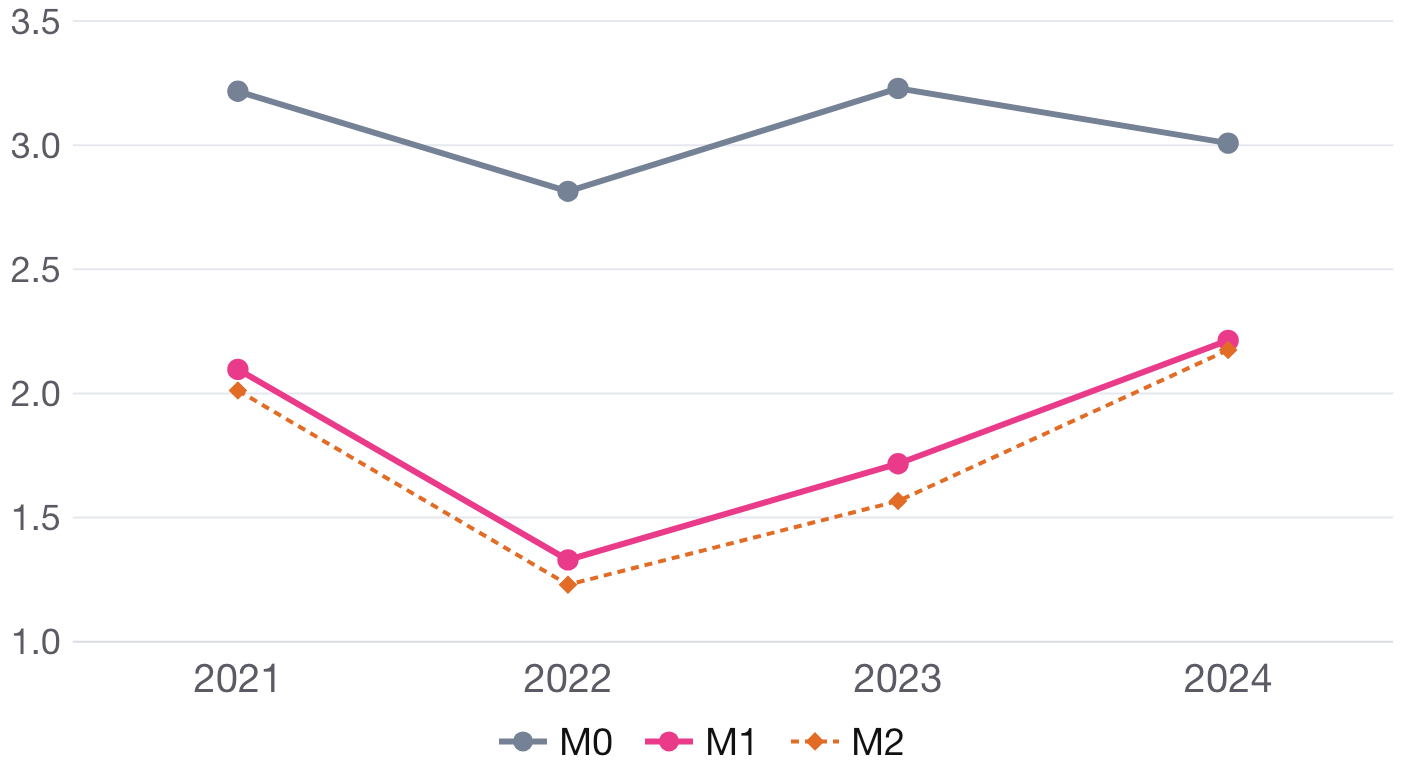
Model	Utility of recipient r for voter v	What changes
M0	$\eta_{vrt} = a_r$	Historical strength of the recipient
M1	$\eta_{vrt} = a_r + \mathbf{x}_{vrt}^{\mathsf{T}}\beta$	14 recipient and pair predictors
M2	$\eta_{vrt} = a_r + \mathbf{x}_{vrt}^{\mathsf{T}}\beta + d_{vr}$	Hierarchical recipient strength + directed pair effects

M2 adds hierarchy and pair effects together; comparison with M1 measures their combined contribution.

C. M2 improves national winner forecasts

2021–2024 validation · 2025 evaluated separately · market predictors included

Categorical log loss · lower is better



National winner · log loss

Model	2021–24 mean	2025
M0	3.067	3.062
M1	1.839	2.143
M2	**1.746**	**2.075**

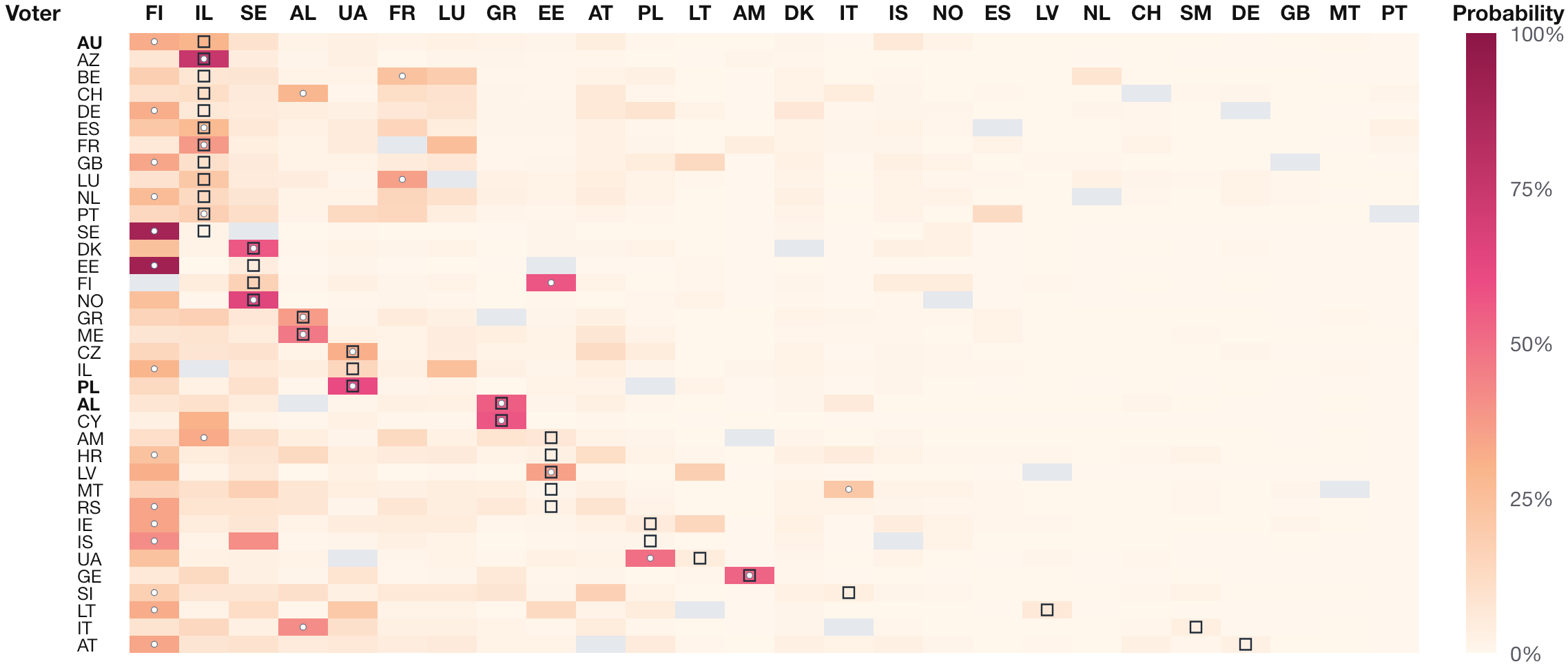
3 / 12 / 14

correct winners out of 36
M0 / M1 / M2 · test 2025

M2 has the lowest log loss in every validation year and in 2025. Four validation years, with equal weight per year.

C. National televote forecasts in 2025

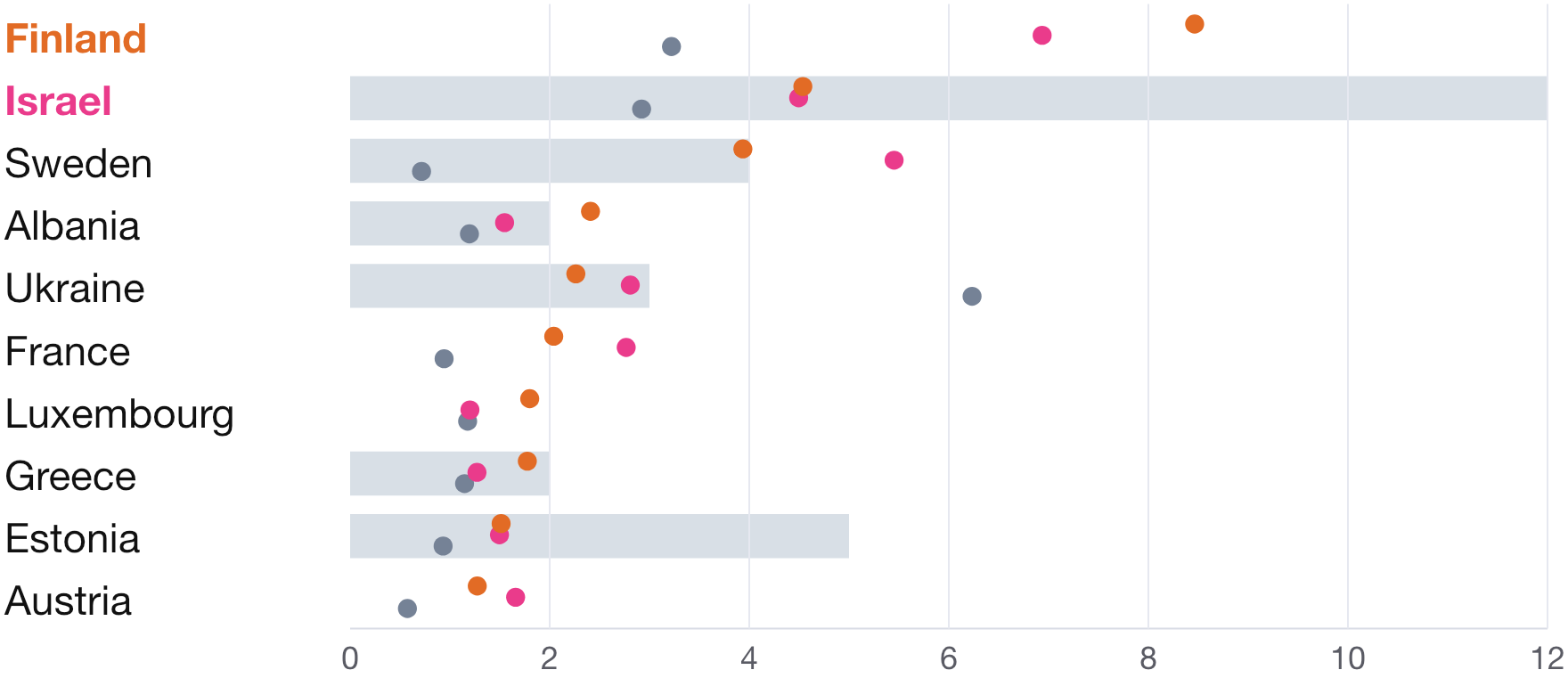
M2 · rows are voters, columns are finalists · 36 genuine national votes



☐ Actual winner ○ M2 favourite Self-vote excluded

C. Where the overall support picture was wrong

2025 · expected versus actual national first places · leading recipients by M2



Finland

8.5 → 0

M2 expected → actual

Israel

4.5 → 12

M2 expected → actual

Expected counts sum win probabilities across voters. These are national first places, not total televote points.

What we could predict

More complex models did not always forecast better

A · Top 10

M1

Predictors helped

M1 and M2 were nearly tied;
both improved on country history.

B · Audience vs jury

M0

Country history was best

Extra predictors did not improve
average forecast scores.

C · National winners

M2

Pairs and hierarchy helped

Best log loss in validation
and in the 2025 test.

Limits of the evidence

Four validation years; retrospective evaluation.
Archived bookmaker odds have no verified timestamp.